

Object Counting Survey Across Modalities

Final 3-Day Codex Computer-Use + Scopus Guide for Collecting 500+ Papers

Prepared for Joana Owusu | Focus: image, video, open-vocabulary/foundation-model counting, 3D, remote sensing, microscopy, thermal, and event-camera counting.

Purpose. This guide is a practical operating plan for using **Codex computer use**, **Codex local coding**, **Google Scholar seed discovery**, **Scopus structured exports**, and scholarly APIs to collect, deduplicate, group, and screen 500+ candidate papers for a journal-level survey on object counting across modalities.

Boundary. Do not ask Codex to scrape Google Scholar. Use Scholar only for supervised seed discovery. Use Scopus only through legitimate institutional access and built-in export tools. Use Codex for organizing files, importing exports, querying scholarly APIs, deduplicating, categorizing, creating BibTeX, and generating reports.

1. Final Goal and Deliverables

The goal is not to deeply read 500 papers in three days. The goal is to build a systematic, defensible corpus: raw records, a cleaned literature matrix, a core reading set, a reproducible query log, and a taxonomy-driven survey outline.

Output	Target	Purpose
seed_papers_google_scholar.xlsx	150-250 high-quality seed papers	Manual Scholar seeds collected with Codex computer-use support.
data/scopus_exports/*.csv or *.bib	500-1,500 structured records before dedup	Recommended bulk source from Scopus structured searches and exports.
master_papers_raw.csv	900-2,000 raw records	All records from Scholar seeds, Scopus exports, APIs, citations, references, and query-bank searches.
literature_matrix_clean.xlsx	500-700 deduplicated candidates	Main working corpus for the survey.
core_reading_set.xlsx	80-150 A-tier papers	Deep reading set for writing the survey.
references_validated.bib	BibTeX for included papers	LaTeX-ready references with unique keys and escaped characters.
screening_log.xlsx	All keep/remove decisions	Transparent record for journal reproducibility.
taxonomy_groups.md + survey_outline_final.md	Writing roadmap	Grouped literature map, suggested sections, tables, figures, and key papers.

2. Source Strategy

Source	Role	How Codex should use it
Google Scholar	Seed discovery and citation chasing	Use supervised browser/computer use only. Inspect top results manually, copy selected metadata, and log exact queries.
Scopus	Structured bulk source	Use advanced search, filters, and export tools. Codex can help run queries after login and import exported CSV/BibTeX files.
Papers With Code	Benchmark/task discovery	Query task/dataset endpoints for object-counting papers and dataset-linked methods.
Semantic Scholar	Citation/reference expansion	Use API/batch endpoints to expand seeds and Scopus records.
OpenAlex	Large-scale metadata enrichment	Resolve DOI/venue/year/abstract/open access links and related works.
Crossref	DOI and publisher metadata	Resolve missing DOI values and improve BibTeX.
arXiv	Recent and gap-fill preprints	Search underrepresented modalities and recent foundation-model/open-vocabulary/video counting papers.

3. Three-Day Timeline

Day	Main goal	End-of-day target
Day 1	Set up project; collect Scholar seeds; run Scopus searches and exports	150-250 Scholar seeds; Scopus CSV/BibTeX files; query log complete.
Day 2	Use Codex to import Scopus and expand with APIs	900-2,000 raw records; 500-700 deduplicated and scored candidates.
Day 3	Screen, gap-fill, validate BibTeX, and generate survey organization	A/B/C tiers, core reading set, taxonomy groups, screening summary, validated BibTeX, final outline.

4. Project Folder and Setup

```

object_counting_survey_codex/
README.md
data/
seed_papers_google_scholar.xlsx
query_log.xlsx
scopus_exports/
scopus_broad.csv
scopus_image_general.csv
scopus_fewshot_exemplar.csv
scopus_openvocab_zeroshot.csv
scopus_video.csv
scopus_3d_rgbd_pointcloud.csv
scopus_remote_sensing_aerial_uav.csv
scopus_medical_microscopy_cell.csv
scopus_benchmarks_datasets.csv
raw_api_results/
master_papers_raw.csv
literature_matrix_clean.xlsx
screening_log.xlsx
scripts/
00_import_scopus_exports.py
01_collect_paperswithcode.py
02_collect_semantic_scholar.py
03_collect_openalex.py
04_collect_crossref.py
05_collect_arxiv.py
06_merge_deduplicate.py
07_score_group_screen.py
08_export_bibtex.py
09_generate_reports.py
outputs/
core_reading_set.xlsx
references.bib
references_validated.bib
taxonomy_groups.md
grouped_papers_final.md
benchmark_compendium.md
collection_summary.md
survey_outline_final.md
papers_pdf/
image/ video/ other_modalities/ datasets/

```

Environment commands

```

python -m venv .venv
source .venv/bin/activate
pip install pandas openpyxl requests tqdm rapidfuzz bibtexparser pyyaml scikit-learn
matplotlib
export S2_API_KEY="your_semantic_scholar_key_here"
export OPENALEX_MAILTO="your_email@domain.com"

```

First Codex prompt

Read this project folder and create any missing directories and template files.
Create README.md describing the 3-day object-counting survey data-collection workflow.
Create data/seed_papers_google_scholar.xlsx, data/query_log.xlsx, and empty output folders.
Do not run API collection yet. Show me the folder plan and ask for approval.

5. Seed Spreadsheet Columns

Column	Purpose
seed_id	Unique ID: S001, S002, ...
paper_title	Full title.

Column	Purpose
authors	Main authors or full author list.
year	Publication year.
venue	CVPR, ICCV, ECCV, NeurIPS, WACV, journal, arXiv, etc.
source_found	Google Scholar, Scopus, reference list, Semantic Scholar, Papers With Code, etc.
google_scholar_query / scopus_query	Exact query used. Use separate fields if both sources apply.
modality	Image; Video; Multimodal; 3D / Point Cloud; RGB-D; Remote Sensing / Aerial; Medical / Microscopy; Thermal; Event Camera; Other.
task_category	Survey; Dataset/Benchmark; Density-based; Detection-based; Few-shot; Class-agnostic; Zero-shot; Open-vocabulary; Foundation-model; Video tracking-based; Application-specific.
input_type	Image; Video; Text Prompt; Exemplar Boxes; Exemplar Patches; Points; Masks; RGB-D/Depth; Point Cloud; Natural Language Query.
output_type	Count Only; Density Map + Count; Boxes + Count; Masks + Count; Points + Count; Tracks + Unique Count; Class-wise Counts; Benchmark.
dataset_or_benchmark	FSC-147, FSCD-147, CountBench, OmniCount, CARPK, ShanghaiTech, TAO-Count, VideoCount, etc.
main_contribution	One-sentence reason this paper matters.
relevance_score	High/Medium/Low at seed stage; later converted to 0-5.
doi / arxiv_id / paper_url	Identifiers and links.
bibtex_collected / pdf_saved / notes	Status fields and observations.

6. Day 1A: Google Scholar Seed Collection with Codex Computer Use

Scholar is used only for seed discovery. For each query, inspect the top 20 relevance-sorted results and top 10 date-sorted results. Save only papers that clearly belong in the survey scope.

Codex prompt for Scholar seed collection

You are helping me collect seed papers for a journal survey on object counting across modalities.
Use browser/computer actions only under my supervision. Do not scrape Google Scholar.

For each query I give you:

1. Inspect only the top 20 relevance-sorted results and top 10 date-sorted results.
2. Save clearly relevant papers into data/seed_papers_google_scholar.xlsx.
3. Fill all available columns: title, authors, year, venue, query, modality, task_category, input_type, output_type, dataset_or_benchmark, DOI/arXiv ID/URL, and notes.
4. Update data/query_log.xlsx with query, date, result count inspected, and papers saved.
5. Stop if Scholar slows down, blocks, or asks for CAPTCHA. I will handle any access prompts.

Scholar query bank

Bucket	Google Scholar queries	Seed target
Survey/taxonomy	"object counting" "survey"; "visual counting" survey; "class-agnostic counting" survey; "few-shot object counting" survey	15-25
Image/few-shot	"few-shot object counting"; "exemplar-based object counting"; "class-agnostic object counting"; "FSC-147" "object counting"	40-60
Zero-shot/open-vocabulary	"zero-shot object counting"; "open-vocabulary object counting"; "text-guided object counting"; "vision-language" "object counting"	35-50
Foundation models	"object counting" "foundation models"; "object counting" SAM; "object counting" CLIP; "visual counting" "vision-language model"	20-35
Video counting	"video object counting"; "object counting in videos"; "unique instance counting" video; CountVid; VideoCount; TAO-Count	35-50
Other modalities	"3D object counting"; "point cloud" "object counting"; "RGB-D object counting"; "thermal" object counting; "event camera" object counting	20-35
Domains/datasets	"remote sensing" "object counting"; "cell counting" "computer vision"; FSC-147; FSCD-147; CountBench; OmniCount; CARPK; ShanghaiTech	50-80

7. Day 1B: Scopus Bulk Collection with Codex Computer Use

Scopus is the recommended source for building a 500+ paper corpus because it supports structured search, filtering, and metadata export. Use Scopus through university access and export records through the interface.

Scopus export fields

Export fields to include when available:
Authors; Title; Year; Source title; Abstract; Author Keywords; Index Keywords;
DOI; Cited by; Document Type; Affiliations; References; Link; EID; ISBN/ISSN.

Scopus search queries

Bucket	Scopus query
Broad object counting	TITLE-ABS-KEY("object counting" OR "visual counting") AND PUBYEAR > 2014
Image object counting	TITLE-ABS-KEY(("object counting" OR "visual counting") AND (image OR images OR "single image")) AND PUBYEAR > 2014
Few-shot / exemplar	TITLE-ABS-KEY("few-shot counting" OR "few shot object counting" OR "exemplar-based counting" OR "exemplar based object counting" OR "class-agnostic counting") AND PUBYEAR > 2014
Zero-shot / open-vocabulary	TITLE-ABS-KEY("zero-shot object counting" OR "zero shot counting" OR "open-vocabulary counting" OR "open vocabulary object counting" OR "text-guided object counting") AND PUBYEAR > 2018
Foundation-model counting	TITLE-ABS-KEY(("object counting" OR "visual counting") AND ("foundation model" OR SAM OR CLIP OR "vision-language model" OR "vision language model")) AND PUBYEAR > 2018
Video counting	TITLE-ABS-KEY("video object counting" OR "object counting in videos" OR "unique object counting" OR "unique instance counting" OR "open-world object counting in videos") AND PUBYEAR > 2014
3D / point cloud / RGB-D	TITLE-ABS-KEY("3D object counting" OR "point cloud counting" OR "RGB-D counting" OR "depth-based counting" OR "occluded object counting") AND PUBYEAR > 2014
Remote sensing / aerial / UAV	TITLE-ABS-KEY(("remote sensing" OR aerial OR UAV OR satellite) AND ("object counting" OR "vehicle counting" OR "building counting" OR "tree counting")) AND PUBYEAR > 2014
Medical / microscopy / cell	TITLE-ABS-KEY(("cell counting" OR "microscopy counting" OR "medical image counting") AND ("deep learning" OR "computer vision")) AND PUBYEAR > 2014
Benchmarks / datasets	TITLE-ABS-KEY("FSC-147" OR "FSCD-147" OR "CountBench" OR "OmniCount" OR "CARPK" OR "ShanghaiTech" OR "TAO-Count" OR "VideoCount")

Codex prompt for Scopus export

You are helping me collect Scopus records for a journal survey.
Use computer/browser actions only under my supervision.
I will handle institutional login, CAPTCHA, and access prompts myself.

For each Scopus query:

1. Paste the query into Scopus Advanced Search.
2. Apply filters where appropriate:
 - Year: 2015-2026 unless keeping older foundational papers.
 - Subject area: Computer Science, Engineering, Medicine/Biomedicine when relevant.
 - Document type: Article, Conference Paper, Review.
 - Language: English.
3. Tell me the result count before export.
4. Export CSV and/or BibTeX with the fields listed in this guide.
5. Save each export inside data/scopus_exports/ using a descriptive filename.
6. Update data/query_log.xlsx with query, date, filters, result count, and export filename.

Do not bypass access restrictions or automate repeated downloads beyond normal Scopus export tools.

8. Taxonomy Rules: Modality, Input Type, Output Type

When to select Multimodal

Select **Multimodal / Vision-Language** when counting combines visual data with another signal: image/video + text prompt, VQA-style question, CLIP/VLM/SAM-style language-guided counting, text + exemplars, RGB-D, image + point cloud, or sensor metadata. Use multiple tags when needed, for example: *Video; Multimodal; Open-Vocabulary*.

Input and output decision rule

Field	Meaning	Examples
input_type	What the method receives before counting	Image; Video; Text Prompt; Exemplar Boxes; Exemplar Patches; Points; Masks; Depth/RGB-D; Point Cloud; Remote Sensing Image; Medical/Microscopy Image; Natural Language Query.
output_type	What the method returns after counting	Count Only; Density Map + Count; Bounding Boxes + Count; Masks + Count; Points + Count; Tracks + Unique Count; Class-wise Counts; Region-wise Counts; Question Answer.

Example labels

Paper type	Modality	Input Type	Output Type
Traditional crowd counting	Image	Image	Density Map + Count
FSC-147 few-shot counting	Image	Image; Exemplar Boxes/Patches	Count Only
Zero-shot text-guided counting	Image; Multimodal	Image; Text Prompt	Count Only or Boxes + Count
SAM/CLIP open-vocabulary counting	Image; Multimodal; Vision-Language	Image; Text Prompt	Masks + Count or Boxes + Count
Video unique counting	Video; possibly Multimodal	Video; Text Prompt or Exemplar	Tracks + Unique Count
3D point-cloud counting	3D / Point Cloud	Point Cloud or RGB-D	Count Only or 3D Boxes + Count

9. Day 2: Codex Expansion, Import, Deduplication, and Scoring

Codex master prompt

I am building a reproducible literature collection pipeline for a journal survey titled: "Object Counting Across Modalities: A Survey of Image, Video, Open-Vocabulary, and Emerging Visual Counting Methods."

You are Codex working in `object_counting_survey_codex/`.
Use local files, Python scripts, Scopus exports, and legitimate scholarly APIs.
Do not scrape Google Scholar.

Inputs:

- `data/seed_papers_google_scholar.xlsx`
- `data/scopus_exports/*.csv` or `*.bib` if available

Build a complete Python pipeline that:

1. Creates missing folders and logs.
2. Imports Scopus exports and standardizes fields.
3. Loads Scholar seed papers.
4. Builds `query_bank.yaml` with modality and task-specific queries.
5. Queries Papers With Code, Semantic Scholar, OpenAlex, Crossref, and arXiv.
6. Expands each seed and Scopus record through references, citations, related papers, dataset searches, and query-bank searches.
7. Saves all raw records to `data/master_papers_raw.csv`.
8. Deduplicates using DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, normalized title, and fuzzy title matching.
9. Assigns modality, method_family, input_type, output_type, application_area, relevance_score, tier, and reason_kept_or_removed.
10. Saves `literature_matrix_clean.xlsx`, `screening_log.xlsx`, `core_reading_set.xlsx`, `references.bib`, `taxonomy_groups.md`, `collection_summary.md`, and `survey_outline_final.md`.
11. Logs every Scholar seed query, Scopus export query, API query, result count, and error.

Before running long API calls, show me the planned scripts and ask for permission.

Codex prompt: import Scopus exports

Read all CSV and BibTeX files in `data/scopus_exports/`.
Create `scripts/00_import_scopus_exports.py` that:

1. Detects Scopus column names and maps them to the project schema.
2. Preserves original Scopus fields in source-specific columns.
3. Adds `source = "Scopus"` and `source_query/export_filename`.
4. Extracts dataset names from title, abstract, author keywords, and index keywords.
5. Assigns initial modality, task_category, input_type, output_type, and application_area.
6. Saves `data/raw_api_results/scopus_imported_records.csv`.
7. Prints counts by export file, year, document type, and preliminary modality.
8. Does not delete duplicates yet; deduplication happens in the master merge step.

Codex prompt: query bank

Create `query_bank.yaml` with these groups:
`survey_taxonomy`, `image_counting`, `few_shot_exemplar`, `zero_shot_open_vocab`,

foundation_models, video_counting, tracking_reid, other_modalities, applications, benchmarks.

Include terms such as:

object counting survey; visual counting survey; few-shot object counting; exemplar-based object counting; class-agnostic counting; zero-shot object counting; open-vocabulary object counting; text-guided object counting; SAM object counting; CLIP object counting; video object counting; unique instance counting video; 3D object counting; point cloud object counting; RGB-D object counting; remote sensing object counting; aerial image object counting; UAV object counting; cell counting computer vision; microscopy counting; FSC-147; FSCD-147; CountBench; OmniCount; CARPK; ShanghaiTech; TAO-Count; VideoCount.

Codex prompt: merge, deduplicate, and score

Read all files in data/raw_api_results/, data/scopus_exports/, and data/seed_papers_google_scholar.xlsx.
Create data/master_papers_raw.csv and data/literature_matrix_clean.xlsx.

Deduplicate using:

- DOI exact match
- arXiv ID exact match
- Semantic Scholar/OpenAlex IDs
- normalized title exact match
- fuzzy title similarity $\geq 92\%$

For each remaining paper, assign:

- modality: image, video, multimodal, 3D, point_cloud, RGB-D, remote_sensing, microscopy, thermal, event_camera, medical, other
- method_family: density_map, regression_based, detection_based, exemplar_based, few_shot, class_agnostic, zero_shot, open_vocabulary, vision_language, segmentation_based, tracking_based, re_identification, foundation_model, dataset_benchmark, survey, application_specific
- input_type and output_type using the taxonomy guide
- application_area
- relevance_score: 0-5
- tier: A_core, B_important, C_background, Exclude
- reason_kept_or_removed: one sentence

Targets:

- Retain 500-700 candidate papers.
- Place 80-150 papers in A_core.
- Put excluded papers in a separate sheet with reasons.
- Print counts by source, modality, method_family, year, and tier.

10. Day 3: Screening, Gap Filling, and Journal-Ready Organization

Coverage targets

Area	Minimum target	Gap-fill query if below target
Image counting	150+	few-shot object counting; class-agnostic counting; dense object counting
Zero-shot/open-vocabulary/foundation models	60+	zero-shot object counting; open-vocabulary object counting; SAM object counting
Video counting	40+	video object counting; unique instance counting; CountVid; VideoCount; TAO-Count
3D / point cloud / RGB-D	20+	3D object counting; point cloud object counting; RGB-D object counting
Remote sensing / aerial / UAV	35+	remote sensing object counting; aerial vehicle counting; UAV object counting
Microscopy / medical / cell	35+	cell counting computer vision; microscopy object counting; medical image counting
Datasets / benchmarks / surveys	50+	FSC-147; FSCD-147; CountBench; OmniCount; TAO-Count; object counting survey

Codex prompt: gap-fill and manual review

Read data/literature_matrix_clean.xlsx and count papers by modality, method_family, dataset, and tier. If any target is below the minimum, run arXiv, Semantic Scholar, OpenAlex, and Scopus-export gap-fill searches if needed.

Create outputs/manual_review_batch.xlsx for uncertain papers where:

- relevance_score is 2 or 3, OR
- modality/method_family is unknown, OR
- abstract is missing but title looks relevant.

For each uncertain paper, add recommended_decision = KEEP, DOWNGRADE, or REMOVE, plus a one-sentence reason. Update screening_log.xlsx after I review the batch.

Codex prompt: final reports and outline

Read the final literature_matrix_clean.xlsx and generate:

1. outputs/taxonomy_groups.md
2. outputs/grouped_papers_final.md
3. outputs/benchmark_compendium.md
4. outputs/survey_outline_final.md
5. outputs/collection_summary.md

For the outline, organize by taxonomy and research gaps, not just chronology. For each major section, include 3-7 key papers, suggested tables/figures, datasets, common metrics, limitations, and open questions.

11. BibTeX and PRISMA-Style Validation

BibTeX validation prompt

Read outputs/references.bib and validate every entry.

1. Escape LaTeX special characters: &, %, \$, #, _.
2. Check balanced braces and duplicate keys.
3. Use key format: FirstAuthorLastname_Year_FirstImportantTitleWord.
4. Required fields:
@article: author, title, journal, year
@inproceedings: author, title, booktitle, year
@misc/arXiv: author, title, year, eprint, archivePrefix
5. Save outputs/references_validated.bib.
6. Create outputs/bibtex_validation_report.md.

PRISMA-style screening prompt

Create outputs/screening_summary.md and data/screening_log.xlsx.

Report:

- Google Scholar seed papers collected manually
- Scopus records exported/imported
- records retrieved from Papers With Code, Semantic Scholar, OpenAlex, Crossref, arXiv
- raw records before deduplication
- duplicates removed
- records screened by title/abstract
- records excluded and top reasons
- papers retained in final 500+ corpus
- A_core, B_important, C_background, Exclude counts
- counts by modality, method_family, year, source, dataset

Also create a simple text PRISMA flow that can become a figure later.

12. Quality Control Rules for Journal Acceptance

Rule	Action
Reproducibility	Record every query, source, date, filters, export filename, API endpoint, result count, kept count, and exclusion reason.
Coverage	Cover images, videos, open-vocabulary/foundation models, and at least three emerging/specialized modalities.
Balance	Do not let crowd counting, generic detection, or one dataset dominate unless that is the stated scope.
Recency	Include strong recent 2022-2026 work, especially zero-shot, open-vocabulary, foundation-model, and video counting.
Foundations	Keep older foundational papers that introduced datasets, metrics, or method families.
Taxonomy	Every included paper must map to modality, method_family, input_type, output_type, and tier.
Manual review	Manually inspect all A-tier papers and uncertain B-tier papers.
Ethical access	No Google Scholar scraping; no access bypassing; Scopus only through legitimate export tools.

13. Final 3-Day Operating Checklist

- Day 1: seed spreadsheet created; 150-250 Scholar seeds collected; Scopus exports saved; query_log.xlsx complete.
- Day 2: Scopus imported; API scripts run; master_papers_raw.csv contains 900-2,000 raw records.
- Day 2: literature_matrix_clean.xlsx contains 500-700 deduplicated candidates with scores and taxonomy labels.
- Day 3: modality gaps filled; A/B/C tiers assigned; 80-150 A-tier papers selected for deep reading.
- Day 3: references_validated.bib, screening_log.xlsx, taxonomy_groups.md, benchmark_compendium.md, collection_summary.md, and survey_outline_final.md generated.
- Before writing: manually review all A-tier papers and questionable B-tier papers.

14. Source Documents Used to Build This Guide

This final guide combines the earlier 500+ paper collection pipeline with the second collection plan that emphasized Papers With Code, Semantic Scholar batch-style expansion, arXiv gap filling, and BibTeX validation. It adds Scopus as the recommended structured bulk source and rewrites the process as a Codex computer-use and local-coding workflow.

Use this document as the operating manual when you open Codex and begin data collection. The strongest workflow is: **Scholar seeds + Scopus exports + API expansion + Codex deduplication/screening + manual A-tier review.**